

Unit 1: Introduction to Semiconductor Physics and Device Technology

Complete Study Notes

1. BAND THEORY AND CLASSIFICATION OF SOLIDS

1.1 Formation of Energy Bands

Key Concept: When atoms combine to form solids, discrete atomic energy levels split into continuous energy bands.

Isolated Atom	Solid Crystal
Discrete energy levels (1s, 2s, 3p)	Energy bands (continuous ranges)
No interaction between atoms	Quantum mechanical interactions
Individual electron states	Pauli exclusion principle applies

Band Formation Process:

- Atoms approach each other in crystal lattice
- Atomic orbitals overlap
- Energy levels split due to:
 - Quantum mechanical interactions
 - Pauli exclusion principle (no two electrons in same quantum state)
- Result: **Valence Band** (lower energy) and **Conduction Band** (higher energy)

1.2 Classification of Solids by Conductivity

Material Type	Band Gap (E _g)	Resistivity (ρ)	Key Feature
Conductors	No gap or overlap	10 ⁻⁸ - 10 ⁻⁴ Ω·m	Partially filled VB or VB-CB overlap
Semiconductors	0.1 - 3 eV	10 ⁻³ - 10 ³ Ω·m	Small band gap
Insulators	> 3 eV	10 ⁸ - 10 ¹⁴ Ω·m	Large band gap

1.3 Conductors (Metals)

Characteristics:

- No significant band gap - electrons move freely
- "Sea of electrons" model - mobile electrons in lattice of positive ions
- High electrical/thermal conductivity
- Temperature effect: Conductivity DECREASES with temperature
 - Higher T → More lattice vibrations (phonons)
 - Phonons scatter electrons → Reduced conductivity

Examples: Cu, Al, Au, Ag, Fe

1.4 Insulators

Characteristics:

- **Very large band gap** ($> 3\text{ eV}$)
- **Valence band:** Completely filled
- **Conduction band:** Completely empty
- **Room temperature thermal energy:** $k_B T \approx 0.025\text{ eV}$ (insufficient to cross gap)
- **Dielectric breakdown:** Strong electric field can force electrons across gap

Examples:

- Diamond ($E_g \approx 5.5\text{ eV}$)
- Glass, rubber, wood, quartz, plastics

1.5 Semiconductors

Key Properties:

- **Moderate band gap:** $0.1 - 3\text{ eV}$
- **At $T = 0\text{ K}$:** Behaves like insulator
- **At room temperature:** Limited conductivity
- **Temperature effect:** Conductivity INCREASES with temperature
 - Higher $T \rightarrow$ More electrons cross band gap
 - More charge carriers available

2. INTRINSIC SEMICONDUCTORS

Definition and Properties

Intrinsic Semiconductor: Pure semiconductor crystal with NO intentional doping

Temperature	Behavior	Carrier Concentration
$T = 0\text{ K}$	Insulator	$n = p = 0$
Room Temperature	Limited conductor	$n = p = n_i$

Key Equations:

- **Charge neutrality:** $n = p$ (electrons = holes)
- **Mass action law:** $np = n_i^2$
- **Intrinsic carrier concentration:** n_i depends on temperature and E_g

Carrier Generation Process:

1. Thermal energy excites electron from VB to CB
2. Creates electron-hole pair (EHP)
3. Both contribute to conduction

Common Examples: Pure Si, Pure Ge

3. EXTRINSIC SEMICONDUCTORS

3.1 n-type Semiconductors

Doping: Add pentavalent atoms (Group V)

- **Dopants:** P, As, Sb (5 valence electrons)
- **Process:** 4 electrons bond with Si, 5th electron loosely bound
- **Result:** Extra electrons easily promoted to CB

Property	Value/Description
Majority carriers	Electrons
Minority carriers	Holes
Donor atoms	N_D (ionized donors)
Electron concentration	$n \approx N_D$
Hole concentration	$p = n_i^2 / N_D$
Fermi level	Shifts UP toward CB

3.2 p-type Semiconductors

Doping: Add trivalent atoms (Group III)

- **Dopants:** B, Ga, In (3 valence electrons)
- **Process:** 3 electrons bond with Si, one bond incomplete (hole)
- **Result:** Holes act as positive charge carriers

Property	Value/Description
Majority carriers	Holes
Minority carriers	Electrons
Acceptor atoms	N_A (ionized acceptors)
Hole concentration	$p \approx N_A$
Electron concentration	$n = n_i^2 / N_A$
Fermi level	Shifts DOWN toward VB

3.3 Charge Neutrality and Mass Action

Fundamental Relations:

1. **Charge Neutrality:** $N_D + p = N_A + n$
2. **Mass Action Law:** $np = n_i^2$ (always true in thermal equilibrium)



4. DIRECT VS INDIRECT BAND GAP

Comparison Table

Feature	Direct Band Gap	Indirect Band Gap
Band structure	CB min and VB max at same k	CB min and VB max at

		different k
Momentum conservation	Direct transition (photon only)	Requires phonon assistance
Light emission efficiency	HIGH - efficient radiative recombination	LOW - inefficient light emission
Examples	GaAs, GaN, InP, CdS	Si, Ge
Applications	LEDs, Lasers, Photodetectors	Logic circuits, Transistors, ICs

5. CARRIER TRANSPORT MECHANISMS

5.1 Drift

Definition: Carrier motion due to electric field

Parameter	Formula	Units
Electron drift velocity	$v_n = \mu_n E$	cm/s
Hole drift velocity	$v_p = \mu_p E$	cm/s
Drift current density	$J = q(n\mu_n + p\mu_p)E$	A/cm ²

5.2 Diffusion

Definition: Carrier motion from high to low concentration

Parameter	Formula	Description
Electron diffusion	$J_n = qD_n (dn/dx)$	Gradient-driven
Hole diffusion	$J_p = -qD_p (dp/dx)$	No external field needed

5.3 Mobility Values

Material	μ_n (cm ² /V·s)	μ_p (cm ² /V·s)	Speed Rating
Si	1400	470	Moderate
Ge	3900	1900	Good
GaAs	8500	400	Excellent (electrons)
InAs	30000	500	Superior (electrons)

Note: Higher mobility → Faster devices, Lower resistive losses

6. P-N JUNCTION DIODE

6.1 Junction Formation

Process Steps:

- 1. **Initial contact:** p-type and n-type regions joined
- 2. **Diffusion:**
 - Holes: p → n
 - Electrons: n → p
- 3. **Recombination:** Carriers meet and annihilate
- 4. **Depletion region formation:** Area depleted of mobile carriers
- 5. **Space charge region:** Fixed ions remain (N⁺ on n-side, N⁻ on p-side)
- 6. **Built-in potential (V_{bi}):**
 - Si: ~0.7 V
 - Ge: ~0.3 V

6.2 Biasing Conditions

Bias Type	Connection	Barrier	Depletion Width	Current
Forward	p(+), n(-)	V _{bi} - V _F	Narrows	Large (I _F)
Reverse	p(-), n(+)	V _{bi} + V _R	Widens	Small (I _s)
Zero	No external	V _{bi}	Equilibrium	Zero

6.3 I-V Characteristics

Shockley Diode Equation:

I = I_s[exp(qV/ηkT) - 1]

Parameters:

- I = Diode current
- I_s = Reverse saturation current (nA to μA range)
- q = Electron charge (1.602 × 10⁻¹⁹ C)
- V = Applied voltage
- η = Ideality factor (1 for Ge, ~2 for Si)
- k = Boltzmann constant (1.38 × 10⁻²³ J/K)
- T = Absolute temperature (K)

6.4 Breakdown Mechanisms

Type	Zener Breakdown	Avalanche Breakdown
Voltage range	< 5-6 V	> 6 V
Doping	Heavy	Light
Depletion width	Very thin	Wide
Mechanism	Quantum tunneling	Impact ionization
Current rise	Sharp	Gradual
Temperature coefficient	Negative	Positive

7. ZENER DIODE

Key Characteristics

Design Features:

- Heavily doped p-n junction
- Very thin depletion region
- Designed to operate in reverse breakdown

Operating Parameters:	Parameter	Symbol	Typical Value	
	Zener voltage	V _z	2.4 - 200 V	
	Minimum current	I _{z(min)}	1-5 mA	
	Maximum current	I _{z(max)}	Device dependent	
	Dynamic resistance	r _z	1-100 Ω	

Applications

1. Voltage Regulation

- Maintains constant output voltage
- Protects against supply variations

2. Surge Protection

- Clamps overvoltages
- Protects sensitive components

3. Waveform Shaping

- Clipping circuits
- Clamping circuits

8. BIPOLAR JUNCTION TRANSISTOR (BJT)

8.1 Structure and Types

Type	Structure	Majority Carriers	Symbol Arrow
NPN	n-p-n	Electrons	Points OUT
PNP	p-n-p	Holes	Points IN

8.2 Terminal Characteristics

Terminal	Doping	Size	Function
Emitter	Heavy (+++)	Small	Injects carriers
Base	Light (+)	Very thin	Controls current
Collector	Moderate (+)	Large	Collects carriers

8.3 Current Relationships

Fundamental Equation: $I_E = I_B + I_C$

Current Gains:

- Common Base (α): $\alpha = I_C/I_E$ (typically 0.95-0.99)

- **Common Emitter (β):** $\beta = I_C/I_B$ (typically 20-300)
- **Relationship:** $\beta = \alpha/(1-\alpha)$

8.4 Operating Modes

Mode	BE Junction	BC Junction	Application	Characteristics
Active	Forward	Reverse	Amplifier	$I_C = \beta I_B$
Saturation	Forward	Forward	Closed switch	$V_{CE} \approx 0.2V$
Cutoff	Reverse	Reverse	Open switch	$I_C \approx 0$
Reverse-Active	Reverse	Forward	Rarely used	-

8.5 Configurations

Configuration	Current Gain	Voltage Gain	Input Z	Output Z
Common Emitter	High (β)	High	Medium	Medium
Common Base	~ 1 (α)	High	Low	High
Common Collector	High ($\beta+1$)	~ 1	High	Low

9. OPTOELECTRONIC DEVICES

9.1 Light Emitting Diode (LED)

Operating Principle:

1. Forward-biased p-n junction
2. Carrier injection and recombination
3. Energy released as photons (direct band gap materials)

Wavelength Calculation:

$$\lambda = hc/E_g = 1.24/E_g \text{ (}\mu\text{m)}$$

Color vs Band Gap:		Color	Wavelength (nm)	Energy (eV)	Material Example	-----				
----- ----- ----- -----					Violet	390-				
420	~3.17	GaN	Blue	450-495	~2.73	GaN	Green	495-570	~2.52	GaP
Yellow	570-590	~2.15	GaAsP	Orange	590-620	~2.08	GaAsP	Red	620-	
770	~1.62	GaAs, AlGaAs								

9.2 Photodiode

Operating Principle:

1. Reverse-biased p-n junction
2. Photons create electron-hole pairs
3. Built-in field separates carriers
4. Generates photocurrent proportional to light intensity

Key Parameters:

- **Responsivity:** A/W (current per unit optical power)

- **Dark current:** Current with no illumination
- **Response time:** Speed of detection

9.3 Solar Cell

Operating Principle:

1. Large-area p-n junction (no external bias)
2. Photovoltaic effect generates voltage/current
3. Maximum power point (MPP) for optimal operation

Efficiency Factors: | **Factor** | **Effect on Efficiency** | |-----|-----
 -----| | **Material** | Mono-Si > Poly-Si > Thin-film | | **Temperature** | Higher T → Lower efficiency | | **Irradiance** | More light → More power | | **Angle** | Perpendicular incidence best | | **Losses** | Series R, shunt R, reflection |

Key Parameters:

- **Open circuit voltage (Voc):** Maximum voltage (no load)
- **Short circuit current (Isc):** Maximum current (shorted)
- **Fill factor (FF):** Measure of "squareness" of I-V curve
- **Efficiency (η):** $P_{out}/P_{in} \times 100\%$

QUICK REFERENCE FORMULAS

Essential Equations

1. **Mass Action Law:** $n_p = n_i^2$
2. **Charge Neutrality:** $N_D^+ + p = N_A^- + n$
3. **Drift Current:** $J = q(n\mu_n + p\mu_p)E$
4. **Shockley Equation:** $I = I_s[\exp(qV/\eta kT) - 1]$
5. **BJT Relations:** $\beta = I_C/I_B$, $\alpha = I_C/I_E$, $\beta = \alpha/(1-\alpha)$
6. **LED Wavelength:** $\lambda = 1.24/E_g$ (μm)
7. **Photon Energy:** $E = h\nu = hc/\lambda$

Important Constants

- **q (electron charge):** 1.602×10^{-19} C
- **k (Boltzmann):** 1.38×10^{-23} J/K
- **h (Planck):** 6.626×10^{-34} J·s
- **Room temp thermal energy:** $kT \approx 0.025$ eV
- **Si band gap:** 1.12 eV
- **Ge band gap:** 0.66 eV
- **GaAs band gap:** 1.43 eV

MCQ TIPS AND COMMON TRICKS

Remember These Key Points:

1. **Temperature Effects:**
 - Metals: R increases with T
 - Semiconductors: R decreases with T
2. **Majority Carriers:**

- n-type: electrons
- p-type: holes

3. BJT Arrow Rule:

- NPN: arrow points OUT
- PNP: arrow points IN

4. Breakdown Voltage:

- Zener: $< 6V$ (tunneling)
- Avalanche: $> 6V$ (impact ionization)

5. Direct vs Indirect:

- LEDs need direct band gap
- Si is indirect (poor light emitter)

6. Typical Values:

- Si V_{bi} : 0.7V
- Ge V_{bi} : 0.3V
- BJT β : 20-300
- BJT $V_{CE(sat)}$: 0.2V

7. Fermi Level Position:

- Intrinsic: Mid-gap
- n-type: Near CB
- p-type: Near VB

8. Biasing for Operation:

- LED: Forward bias
- Photodiode: Reverse bias
- Zener: Reverse bias
- Solar cell: No external bias